Question 1

For each of the following functions determine whether it is convex, concave, quasiconvex, or quasiconcave (you should learn the defs of quasiconvex, and quasiconcave by yourself).

(a) $f(x) = e^x - 1$ on $\mathbf{R}$.

$f''(x) = e^x > 0 \Rightarrow f(x)$ is convex.

(b) $f(x_1, x_2) = x_1 x_2$ on $\mathbf{R}^2_{++}$.

1) $\nabla^2 f(x_1, x_2) = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$. the eigenvalues are -1 and 1. $\begin{cases} \text{not PSD} \Rightarrow \text{not convex} \\ \text{not NSD} \Rightarrow \text{not concave} \end{cases}$

2) $\mathrm{dom}(f)$ is obviously convex.

for $\forall \alpha$. $f(x_1, x_2)$'s sublevel set $S_\alpha = \{x \in \mathrm{dom}(f) \mid x_1 x_2 \le \alpha\}$

let $A = (2, 0.5) \in S_1$, $B = (0.5, 2) \in S_1$.

$C = \frac{A+B}{2} = (1.25, 1.25)$, $C \notin S_1 \Rightarrow$ not a convex set

$\Rightarrow f(x_1, x_2)$ not quasiconvex.

3) for f. its superlevel set $S_\beta = \{x \in \mathrm{dom}(f) \mid x_1 x_2 \ge \beta\}$

when $\beta \le 0 \Rightarrow x_1 x_2 \ge \beta$ is always true. $\Rightarrow S_\beta$ is convex.

when $\beta > 0 \Rightarrow x_1 x_2 \ge \beta \Rightarrow x_1 \ge \frac{\beta}{x_2}$.

let $\phi(x) = \frac{\beta}{x}$ $(x > 0)$. $\nabla^2 \phi(x) = \frac{2\beta}{x^3} > 0 \Rightarrow \phi(x)$ is convex

$\Rightarrow$ the epigraph is a convex set

$\Rightarrow S_\beta$ is convex when $\beta > 0$

$\Rightarrow f$ is a quasiconcave function

(c) $f(x_1, x_2) = 1/(x_1 x_2)$ on $\mathbf{R}^2_{++}$.

$\nabla^2 f(x_1, x_2) = \begin{bmatrix} \frac{2}{x_1^3 x_2} & \frac{1}{x_1^2 x_2^2} \\ \frac{1}{x_1^2 x_2^2} & \frac{2}{x_1 x_2^3} \end{bmatrix} = \frac{4}{(x_1 x_2)^4} - \frac{1}{(x_1 x_2)^4} = \frac{3}{x_1^4 x_2^4} > 0$, $\frac{2}{x_1^3 x_2} > 0$

$\Rightarrow \nabla^2 f$ is PSD $\Rightarrow$ convex

(d) $f(x_1, x_2) = x_1/x_2$ on $\mathbf{R}^2_{++}$.

1) $\nabla^2 f(x_1, x_2) = \begin{bmatrix} 0 & \frac{-1}{x_2^2} \\ \frac{-1}{x_2^2} & \frac{2x_1}{x_2^3} \end{bmatrix}$, $|\nabla^2 f(x_1, x_2)| = -\frac{1}{x_2^4} < 0 \Rightarrow f(x_1, x_2)$ neither convex nor concave.

2) let S_α be $f(x_1, x_2)$'s sublevel sets, $S_\alpha = \{x \in \mathrm{dom}(f) \mid x_1/x_2 \le \alpha\}$

$x_1/x_2 \le \alpha \Rightarrow x_1 \le \alpha x_2 \Rightarrow x_1 - \alpha x_2 \le 0 \Rightarrow (1, -\alpha)^T \cdot x \le 0$

$\Rightarrow$ halfspaces are convex $\Rightarrow S_\alpha$ is convex $\Rightarrow f$ is quasiconvex

3) let S_β be $f(x_1, x_2)$'s superlevel sets. $S_\beta = \{x \in \mathrm{dom}(f) \mid x_1/x_2 \ge \beta\}$

$\Rightarrow$ similarly, we can prove S_β's convexity $\Rightarrow f$ is quasiconcave

(e) $f(x_1, x_2) = x_1^2/x_2$ on $\mathbf{R} \times \mathbf{R}_{++}$.

$\nabla^2 f(x_1, x_2) = \begin{bmatrix} \frac{2}{x_2} & \frac{-2x_1}{x_2^2} \\ \frac{-2x_1}{x_2^2} & \frac{2x_1^2}{x_2^3} \end{bmatrix}$ $|\nabla^2 f(x_1, x_2)| = \frac{4x_1^2}{x_2^4} - \frac{4x_1^2}{x_2^4} = 0$, $\frac{2}{x_2} > 0$

$\Rightarrow f(x_1, x_2)$ is convex

(f) $f(x_1, x_2) = x_1^\alpha x_2^{1-\alpha}$, where $0 \le \alpha \le 1$, on $\mathbf{R}^2_{++}$.

$\nabla^2 f(x_1, x_2) = \begin{bmatrix} \alpha(\alpha-1) \cdot x_1^{\alpha-2} x_2^{1-\alpha}, & \alpha(1-\alpha) x_1^{\alpha-1} \cdot x_2^{-\alpha} \\ \alpha(1-\alpha) \cdot x_1^{\alpha-1} \cdot x_2^{-\alpha}, & \alpha(\alpha-1) \cdot x_1^{\alpha} \cdot x_2^{-\alpha-1} \end{bmatrix}$

$|\nabla^2 f| = \alpha^2(\alpha-1)^2 \cdot x_1^{2\alpha-2} \cdot x_2^{-2\alpha} - \alpha^2(\alpha-1)^2 \cdot x_1^{2\alpha-2} \cdot x_2^{-2\alpha} = 0$

when $0 < \alpha < 1$, $\alpha(\alpha-1) x_1^{\alpha-2} x_2^{1-\alpha} < 0$

$\Rightarrow \nabla^2 f(x_1, x_2)$ NSD $\Rightarrow$ concave.

when $\alpha = 0$, $f(x_1, x_2) = x_2 \Rightarrow$ concave & convex

when $\alpha = 1$ $f(x_1, x_2) = x_1 \Rightarrow$ concave & convex

Question 2

Prove that $f(X) = \mathbf{tr}\,(X^{-1})$ is convex on $\mathbf{dom} f = \mathbf{S}^n_{++}$.

Since $X \in S^n_{++}$, let $X = Q\Lambda Q^T = Q\Lambda Q^{-1}$

let $X = Z + tV$, $(t \in R, Z > 0, V \in S^n_{++})$

$\Rightarrow g(t) = f(Z+tV) = tr[(Z+tV)^{-1}] = tr[Z^{-\frac{1}{2}}(I + Z^{-\frac{1}{2}} tV Z^{-\frac{1}{2}})^{-1} \cdot Z^{-\frac{1}{2}}]$

$= tr[Z^{-1}(I + Z^{-\frac{1}{2}} tV Z^{-\frac{1}{2}})^{-1}]$

let $Z^{-\frac{1}{2}} V Z^{-\frac{1}{2}} = Q\Lambda Q^T$.

$\Rightarrow g(t) = tr[Z^{-1} \cdot Q (I + t\cdot\Lambda)^{-1} Q^T]$

$= tr[QZ^{-1}Q^T (I + t\cdot\Lambda)^{-1}]$

$= \sum_{i=1}^{n} (QZ^{-1}Q^T)_{ii} \cdot (1 + t\lambda_i)^{-1}$

for $\forall i$, $g_i(t) = (QZ^{-1}Q^T)_{ii} \cdot (1+t\lambda_i)^{-1}$

$g_i''(t) = \dfrac{2(QZ^{-1}Q^T)_{ii} \cdot \lambda_i^2}{(1+t\lambda_i)^2} > 0 \Rightarrow g_i(t)$ is convex.

$\Rightarrow$ As sum of convex functions, $g(t)$ is convex

$\Rightarrow f(X)$ is convex.

Question 3

Show that for $p > 1$,

$$f(x,t) = \frac{|x_1|^p + \cdots + |x_n|^p}{t^{p-1}} = \frac{\|x\|_p^p}{t^{p-1}}$$

is convex on $\{(x,t) | t > 0\}$.

Introduce $h(x)$, where $f(x,t) = t \cdot h(\frac{x}{t})$. $\mathrm{dom} f = \{(x,t) \mid \frac{x}{t} \in \mathrm{dom}(h), t > 0\}$

$\Rightarrow t \cdot h(\frac{x}{t}) = \frac{\|x\|_p^p}{t^{p-1}} \Rightarrow h(\frac{x}{t}) = \frac{\|x\|_p^p}{t^p} = \left(\frac{\|x\|_p}{t}\right)^p = \|\frac{x}{t}\|_p^p$

$\Rightarrow h(u) = \|u\|_p^p$

when $p > 1$. let $a(x) = x^p$, $b(x) = \|x\|_p$.

$\Rightarrow h = a \circ b$.

obviously $a(x), b(x)$ are all convex.

a's extended-value extension $\tilde{a} = a(x)$ non-decreasing

$\Rightarrow h$ is convex. $\Rightarrow$ h's perspective function f is convex.

Question 4

Derive the conjugates of the following functions.

(a) *Max function.* $f(x) = \max_{i=1,\ldots,n} x_i$ on $\mathbf{R}^n$.
(try $n = 2, 3$ first, then prove it).

(b) *Piecewise-linear function on* $\mathbf{R}$. $f(x) = \max_{i=1,\ldots,m}(a_i x + b_i)$ on $\mathbf{R}$. You can assume that the a_i are sorted in increasing order, *i.e.*, $a_1 \le \cdots \le a_m$, and that none of the functions $a_i x + b_i$ is redundant, *i.e.*, for each k there is at least one x with $f(x) = a_k x + b_k$.

(a). $f^*(y) = \sup_{x \in R^n} (y^T x - f(x)) = \sup_{x \in R^n} (y^T x - \max_i x_i)$

let $\max_i x_i = x_a$, $x_a \not\to +\infty$.

let x_b, where $b \neq a$, $x_b \to -\infty$,

$\Rightarrow$ when $y_b < 0$, $y_b \cdot x_b \to +\infty \Rightarrow f^*(y) = +\infty$

$\Rightarrow$ when any $y_k < 0$, $f^*(y) = +\infty$.

when $y \geq 0$. $f^*(y) = \sup_{x \in R} (y^T x - \max_i x_i)$

suppose $x = a \cdot \mathbb{1}^n \Rightarrow y^T x - \max_i x_i = a\sum_{i=1}^{n} y_i - a = a(\sum_{i=1}^{n} y_i - 1)$

$= a(y^T \mathbb{1}^n - 1)$

when $y^T \mathbb{1}^n - 1 \neq 0$, $\sup(y^T x - \max_i x_i)$ can reach $+\infty$ by corresponding manipulation of a.

$\Rightarrow y \geq 0$ & $\mathbb{1}^T y = 1$

$y^T x = \sum_i x_i y_i \le (\sum_i y_i) \max_i x_i$

$= 1 \cdot \max_i x_i$

$= \max_i x_i$

$\Rightarrow f^*(y) = \begin{cases} 0 & y \geq 0 \;\&\; \mathbb{1}^T y = 1 \\ +\infty & \text{otherwise.} \end{cases}$

(b) *Piecewise-linear function on* $\mathbf{R}$. $f(x) = \max_{i=1,\ldots,m}(a_i x + b_i)$ on $\mathbf{R}$. You can assume that the a_i are sorted in increasing order, *i.e.*, $a_1 \le \cdots \le a_m$, and that none of the functions $a_i x + b_i$ is redundant, *i.e.*, for each k there is at least one x with $f(x) = a_k x + b_k$.

$$f^*(y) = \sup_{x \in R} [yx - f(x)]$$

since each line is active. define the breakpoints x_k by:

$$x_k = \frac{b_{k+1} - b_k}{a_k - a_{k+1}}$$

$$\Rightarrow f^*(y) = \begin{cases} +\infty, & y < a_1 \text{ or } y > a_m, \\ y x_k - (a_k x_k + b_k), & y \in [a_k, a_{k+1}], \quad k = 1, \ldots, m-1 \end{cases}$$

Question 5

Show that the conjugate of $f(X) = \mathbf{tr}(X^{-1})$ with $\mathbf{dom} f = \mathbf{S}^n_{++}$ is given by

$$f^*(Y) = -2\mathbf{tr}(-Y)^{1/2}, \quad \mathbf{dom} f^* = -\mathbf{S}^n_{+}.$$

Hint. The gradient of f is $\nabla f(X) = -X^{-2}$.

if $Y \notin -S^n_+$ · then $f^*(Y) = +\infty$, Then Y has an eigenvector q_1.

Take an eigen decomposition $Y = Q \Lambda Q^T$ and chose

$X = Q \operatorname{diag}(t, 1, \ldots, 1)$, $t \to +\infty$.

Then $tr(XY) = t\lambda_1 + \sum_{i=2}^{n} \lambda_i$, $tr(X^{-1}) = \frac{1}{t} + (n-1)$

so $tr(XY) - tr(X^{-1}) \to +\infty$, Hence $f^*(Y) = +\infty$ and $\operatorname{dom}(f^*) \subseteq S^n_+$

if $Y \in -S^n_{++}$, then $f^*(Y) = -2tr[(-Y)^{\frac{1}{2}}]$

For $X \in S^n_{++}$, $\phi(X) = tr(XY) - tr(X^{-1})$ is concave

$\nabla tr(XY) = Y$, & $\nabla tr(X^{-1}) = -X^{-2}$, the first order condition is

$0 = \nabla \phi(X) = Y + X^{-2} \Rightarrow X = (-Y)^{-\frac{1}{2}}$

$\Rightarrow f^*(Y) = tr[(-Y)^{-\frac{1}{2}} Y] - tr\{[(-Y)^{-\frac{1}{2}}]^{-1}\} = -2tr[(-Y)^{\frac{1}{2}}]$

$$\Rightarrow f^*(Y) = \begin{cases} -2\, tr[(-Y)^{\frac{1}{2}}], & Y \in -S^n_{++} \\ +\infty, & Y \in -S^n_+ \end{cases}$$

Question 6

Conjugate of negative normalized entropy. Show that the conjugate of the negative normalized entropy

$$f(x) = \sum_{i=1}^{n} x_i \log(x_i / \mathbf{1}^T x),$$

with $\mathbf{dom} f = \mathbf{R}^n_{++}$ is given by

$$f^*(y) = \begin{cases} 0 & \sum_{i=1}^{n} e^{y_i} \le 1 \\ +\infty & \text{otherwise.} \end{cases}$$

$$f^*(y) = \sup_x (y^T x - f(x))$$

Let $s = \mathbb{1}^T x > 0$ and $p = \frac{x}{s}$, $\Rightarrow p \in \underbrace{\{p \ge 0, \mathbb{1}^T p = 1\}}_{\mathbb{P}}$ and $x = sp$.

Then, $f(x) = \sum_i s p_i \log p_i = s \sum_i p_i \log p_i$.

$\Rightarrow y^T x - f(x) = s(\sum_i p_i y_i - \sum_i p_i \log p_i)$

$$\Rightarrow f^*(y) = \begin{cases} 0, & \sup_{p \in \mathbb{P}} (\sum_i p_i y_i - \sum_i p_i \log p_i) \le 0 \\ +\infty, & \text{other wise} \end{cases}$$

$\sup_{p \in \mathbb{P}} (\sum_i p_i y_i - \sum_i p_i \log p_i) = \log(\sum_{i=1}^{n} e^{y_i})$, attained at $p_i = \frac{e^{y_i}}{\sum_i e^{y_i}}$

$\Rightarrow \log(\sum_i e^{y_i}) \le 0 \Leftrightarrow \sum_i e^{y_i} \le 1$

$$\Rightarrow f^*(y) = \begin{cases} 0 & \sum_i e^{y_i} \le 1 \\ +\infty, & \text{otherwise.} \end{cases}$$